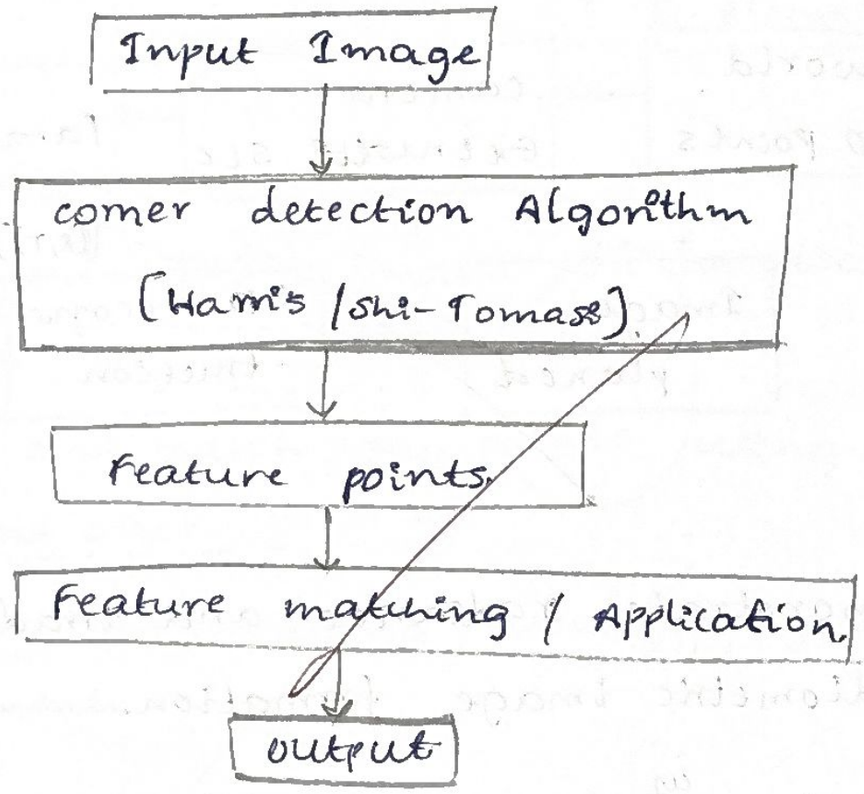
Home Assignment - 02

CO-02

① Explain corner detection algorithms for feature extraction.
6M



Explanation:-

+ Detect interest points using Harris, Shi-Tomasi or FAST.

+ Corners are robust features for tracking and matching

+ Used in face recognition and panorama stecching.

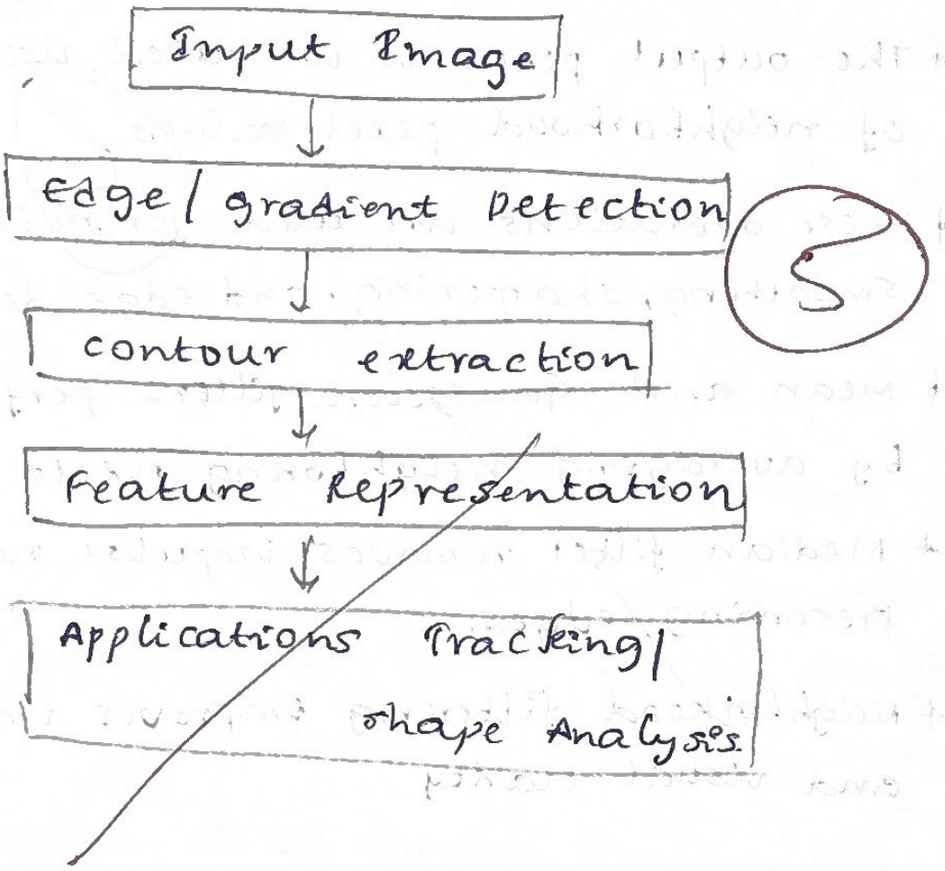
+ Extracted corners are compared accross images for allignment / recognition.

+ Helps in reducing computation by focusing only for key points

+ corner detection is more reliable than edge only features under varying illuminat-ion.

6

(2) Apply contour-based feature ditection for
5M object boundaries.



5

+ Detect edges using gradient / threshold-methods
+ Extract contours representing object.
+ compute featured like shape boundaries descriptors / moments
+ useful in object tracking, shape analysis and classification.
+ contour-based detection handles compare shapes effectively
+ can be combined with motion analysis

Vedio

③ Explain neighborhood filtering operations

5M

+ Neighborhood filtering operations modify a pixel value using its surrounding pixels.

+ A small window (such as 3x3 or 5x5) is moved across the image.

+ The output pixel is computed using a function of neighborhood pixel values.

+ These operations are used for noise reduction, smoothing, sharpening, and edge detection.

+ Mean and Gaussian filters perform smoothing by averaging neighboring pixels

+ Median filter removes impulse noise while preserving edges.

+ Neighborhood filtering improves image quality and visual clarity

P_1	P_2	P_3
P_4	P_5	P_6
P_7	P_8	P_9

→ New pixel $= f(P_1 - P_9)$

(or)

Noisy Image → Neighborhood Filter → Filtered Image.

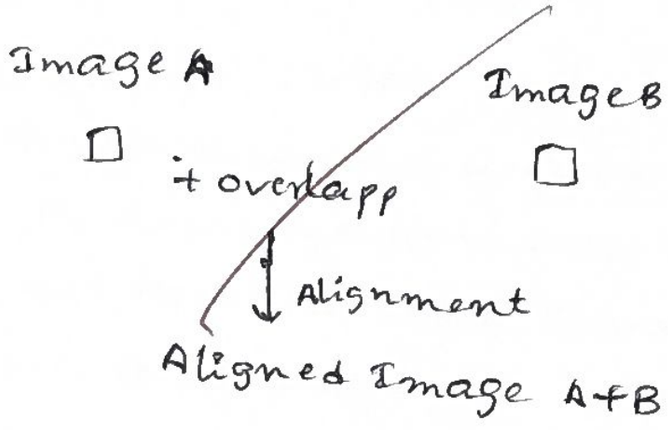
④ Explain the concept of pairwise image alignment

2M

Pairwise image alignment aligns two overlapping images by matching corresponding features and estimating a geometric transformation so that both images are in the same reference

frame.

real-world Eg:- Panaroma creation with a phone without pairwise alignment → misalignment & ghosting



(5) Explain the image acquisition pipeline.
2M

- The image acquisition pipeline converts a real-world scene into a digital image through sensing, sampling, and quantization.
- It includes illumination, lens focusing, sensor capture, and digitization.

Every photo you take follows this pipeline

Scene
↓
Lens
↓
Sensor → Sampling → Quantization
↓
Digital Image

Very

18/20